Inventory Reorder Planner

What you need to build

A shop predicts stockouts and prepares purchase requests without over-ordering.

This is the demo target specified in your official catalogue entry. Build and test against it. If a live service blocks access or requires a paid plan, ask ZIO or the organizer how to demonstrate the permitted workflow.

How your agent should work

01

Collect the details

Stock levels, sales history, lead time and budget.

02

Plan the next steps

Forecasts demand during lead time minus incoming stock.

03

Find the information

Inventory and supplier order dashboards.

04

Take action

Human approval required

Prepares reorder requests for approval.

What your agent should produce

Stockout forecast, proposed quantities and request IDs.

How to check it works

Checks SKU, pack size and existing open orders.

Step-by-step build guide

Implementation suggestions to help you understand the official workflow above. These suggestions do not add judging requirements.

1. Set up the Inventory Reorder Planner inputs

Create a sample record or form for: Stock levels, sales history, lead time and budget. Give every item a stable ID and validate these fields before running the agent.

2. Connect the named sources

Retrieve from: Inventory and supplier order dashboards. Build against Organiser mock inventory dashboard. Store source IDs/URLs and retrieval times beside the extracted values so the decisions can be checked.

3. Implement this problem's decision rules

Convert this rule into explicit comparisons, classifications or model outputs: Forecasts demand during lead time minus incoming stock. Save the input evidence and reason for each decision; return an uncertain state when required evidence is missing.

4. Implement the permitted action

Action to demonstrate: Prepares reorder requests for approval. Show the proposed destination and change for approval before executing it. Store the approved version and prevent duplicate execution on retries.

5. Check the actual result

Implement this verification separately from the action: Checks SKU, pack size and existing open orders. Capture the observed state and compare it with what was intended; keep failed or pending results visible.

6. Build this problem's result view

Arrange the evidence and action states into: Stockout forecast, proposed quantities and request IDs. Make each result traceable to its input record and show any unresolved item.

7. Run the Inventory Reorder Planner scenarios

Create 4 labelled runs, one for each official test case below. For every run, preserve its specified input, expected result, actual result and evidence; explain failures or differences instead of presenting them as success.

Suggested tools for this problem

Optional choices, not additional judging requirements. Use equivalent tools you know. Follow the official demo target; a paid service or real customer account is not required unless the official brief explicitly says so.

HTTP Request + Cheerio; Playwright if needed

Retrieve the brief's sources: Inventory and supplier order dashboards. Use the permitted demo target and retain record URLs or IDs.

SQLite / Supabase + a simple web interface

Store Inventory Reorder Planner inputs, source evidence, action states and test results; show approval controls when the brief requires them.

Official test cases to demonstrate

Run each scenario and capture the input, actual result and evidence. Explain any difference from the expected result.

TC1 | Scenario: Low stock with no incoming order

Expected: correct reorder quantity

TC2 | Scenario: Supplier sells in cases of 12

Expected: converted to units

TC3 | Scenario: Pending order exists

Expected: no duplicate reorder

TC4 | Scenario: Supplier price jumps 20%

Expected: budget review requested

Demo checklist & safety rules

Plain-English guidance from the official brief. Use it together with the workflow and official test cases above.

Run every official test case above and show the input, result and evidence for each.
Ask for human approval before any action that cannot easily be undone.
Use mock or synthetic data when a workflow involves personal or sensitive information.
Do not bypass OTP or CAPTCHA. Show a safe manual step or a mock instead.

How your project is reviewed

The same criteria apply to solo builders and teams.

01
Problem fit
Your agent is assessed against your assigned problem and industry.

02
Working evidence
Organizers check your demo, source and submitted evidence.

03
Jury selection
Organizers confirm the shortlist for jury review.

Scoring criteria · 100 points

Problem fit & usefulness	**20**
Agentic capability	**25**
Technical implementation	**20**
Innovation	**15**
User experience	**10**
Presentation	**10**